UNIT-III

MAGNETOSTATICS:

The no. of electric field lines passing through a closed surface is called Electric flux.

Lorentz force:

The Electric force is given by

$$\vec{F_e} = q\vec{E}$$

The Magnetic force is given by

$$\vec{F_m} = q(\vec{V} \times \vec{B})$$

$V \rightarrow$ Velocity of charge

$\vec{B} \rightarrow$ Magnetic flux density.

Lorentz Law states that:

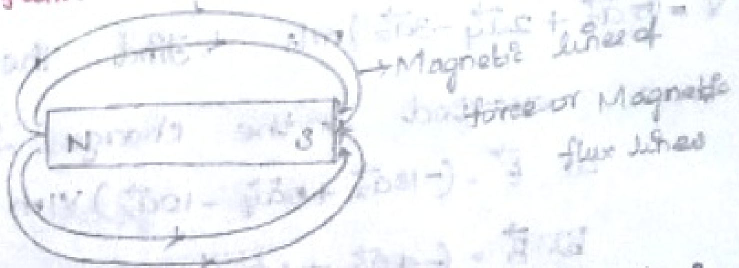
$$\vec{F} = \vec{F_e} + \vec{F_m}$$

$$\vec{F} = q\vec{E} + q(\vec{V} \times \vec{B}) \quad \text{Newton's first Law.}$$

$$m\vec{a} = q\vec{E} + q(\vec{V} \times \vec{B}) \qquad F = m \times a, \quad F = m\vec{a}$$

↳ Lorentz equation.

Magnetic flux:



→ The number of magnetic lines of force passing through a surface is called magnetic flux

→ Denoted by ϕ.

→ Unit → wb or weber

Magnetic flux density:

The number of magnetic lines of force or magnetic flux lines passing through a unit surface.

$B = \phi/s$

$B \rightarrow$ Magnetic flux density.

$\phi = \oint_s \vec{B} \cdot \vec{ds} = 0 \rightarrow$ Gauss law for Magnetostatic

Unit: Wb/m2 or Tesla.

$\nabla \cdot \vec{B} = 0$ (Maxwell Equation)

Magnetic field intensity:

$\vec{H} = F/\phi.$

Unit: N/wb.

Relation b/w $\vec{B}$ and $\vec{H}$:

$\vec{B} = \mu \vec{H}.$

$\mu \rightarrow$ permeability.

---

1. A point charge of $q = -1.2C$ has velocity $\vec{V} = (5\vec{a_x} + 2\vec{a_y} - 3\vec{a_z})$ m/s. Find the magnetic force exerted on the charge if

a). $\vec{E} = (-18\vec{a_x} + 5\vec{a_y} - 10\vec{a_z})$ V/m.

b). $\vec{B} = (-4\vec{a_x} + 4\vec{a_y} + 3\vec{a_z})$ T

c). Assume that both $\vec{E}$ and $\vec{B}$ are simultaneously present.

Soln:

1) $\vec{F_e} = q\vec{E}$

$= -1.2(-18\vec{a_x} + 5\vec{a_y} - 10\vec{a_z})$

$\vec{F_e} = 21.6\vec{a_x} - 6\vec{a_y} + 12\vec{a_z}$ N

2) $\vec{F_m} = q(\vec{V} \times \vec{B})$

$$\vec{V} \times \vec{B} = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ 5 & 2 & -3 \\ -4 & 4 & 3 \end{vmatrix}$$

$$= \vec{a_x}(6+12) - \vec{a_y}(15-12) + \vec{a_z}(20+8)$$

$$= 18\vec{a_x} - 3\vec{a_y} + 28\vec{a_z}$$

$$\vec{F_m} = -1.2(18\vec{a_x} - 3\vec{a_y} + 28\vec{a_z})$$

$$\vec{F_m} = -21.6\vec{a_x} + 3.6\vec{a_y} - 33.6\vec{a_z} \ N$$

3) $\vec{F} = \vec{F_e} + \vec{F_m}$

$$\vec{F} = -2.4\vec{a_y} - 21.6\vec{a_z} \ N \qquad \boxed{|F| = 21.73 N}$$

Ampere circuital Law refer back

Scalar Magnetic Potential:

The Scalar Magnetic Potential (Vm) is expressed as

$$V_m = -\int_a^b \vec{H} \cdot \vec{dl} \longrightarrow ①$$

The following vector identity is used to define scalar magnetic potential.

$$\nabla \times (\nabla V) = 0 \longrightarrow ②$$

If 'Vm' is scalar magnetic potential, it must satisfy equ ②

$$\nabla \times (\nabla V_m) = 0 \longrightarrow ③$$

We can relate 'Vm' with $\vec{H}$ as.

$$\vec{H} = -\nabla V_m$$

$$\nabla V_m = -\vec{H} \longrightarrow ④$$

sub equ ④ in equ ③

$$\nabla \times (-\vec{H}) = 0$$

$$\nabla \times \vec{H} = 0 \rightarrow (5)$$

But by point form of Ampere's Law,

$$\nabla \times \vec{H} = \vec{J} \rightarrow (6)$$

compare equ (5) and (6)

$$\vec{J} = 0$$

Hence, scalar magnetic potential 'Vm' can be defined only for source free region where $\vec{J} = 0$

$$\vec{H} = -\nabla V_m \text{ only for } \vec{J} = 0.$$

Laplace Equation for 'Vm':

For a closed surface, in Magnetostatics

$$\oint \vec{B} \cdot d\vec{s} = 0 \rightarrow (7)$$

By divergence theorem, equ (7) can be rewritten as

$$\int (\nabla \cdot \vec{B})\, dv = 0$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \cdot (\mu \vec{H}) = 0$$

$$\nabla \cdot \vec{H} = 0$$

$$\nabla \cdot (-\nabla V_m) = 0$$

$$\nabla^2 V_m = 0, \text{ only if } \vec{J} = 0.$$

Laplace equation for scalar magnetic potential is

$$\nabla^2 V_m = 0$$

Vector Magnetic Potential ($\vec{A}$):

It is defined as a quantity whose curl gives the Magnetic flux density.

$$\vec{B} = \nabla \times \vec{A} \longrightarrow ① \; ; \vec{J} \neq 0$$

Where

$\vec{A} \rightarrow$ Vector magnetic potential

Unit - Ampere

$$\vec{A} = \frac{\mu}{4\pi} \int_V \left(\frac{\vec{J}}{r}\right) dv \longrightarrow ②$$

The following Vector identity is used to define Vector magnetic potential.

$$\nabla \cdot (\nabla \times \vec{A}) = 0 \longrightarrow ③$$

If $\vec{A}$ is a Vector magnetic potential, it must satisfy the equ ③

We know, Gauss Law in point form:

$$\nabla \cdot \vec{B} = 0 \longrightarrow ④$$

compare equ ③ and ④

$$\vec{B} = \nabla \times \vec{A}$$

By point form of Ampere's Law,

$$\nabla \times \vec{H} = \vec{J}$$

$$\nabla \times \left(\frac{\vec{B}}{\mu}\right) = \vec{J}$$

$$\nabla \times \vec{B} = \mu \vec{J} \longrightarrow ⑤$$

Sub equ ① in equ ⑤

$$\nabla \times (\nabla \times \vec{A}) = \mu \vec{J}$$

By Vector Triple product,

$$\nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = \mu \vec{J}$$

For the time Varying field

$$\nabla \cdot \vec{A} = 0$$

$$-\nabla^2 \vec{A} = \mu \vec{J}$$

$$\nabla^2 \vec{A} = -\mu \vec{J}$$

$\downarrow$

Poisson equation for magnetic field.

$$\vec{A} = A_x \vec{a_x} + A_y \vec{a_y} + A_z \vec{a_z}$$

$$\vec{J} = J_x \vec{a_x} + J_y \vec{a_y} + J_z \vec{a_z}$$

$$\nabla^2 A_x \vec{a_x} + \nabla^2 A_y \vec{a_y} + \nabla^2 A_z \vec{a_z} = -\mu (J_x \vec{a_x} + J_y \vec{a_y} + J_z \vec{a_z}).$$

$$\left.\begin{aligned} \nabla^2 A_x &= -\mu J_x \\ \nabla^2 A_y &= -\mu J_y \\ \nabla^2 A_z &= -\mu J_z \end{aligned}\right\} \text{in the form of Poisson's equation.}$$

The magnetic potential can be written as

$$\vec{A_x} = \frac{\mu}{4\pi} \int_V \left(\frac{\vec{J_x}}{r}\right) dv$$

$$\vec{A_y} = \frac{\mu}{4\pi} \int_V \left(\vec{J_y}/r\right) dv$$

$$\vec{A_z} = \frac{\mu}{4\pi} \int_V \left(\vec{J_z}/r\right) dv.$$

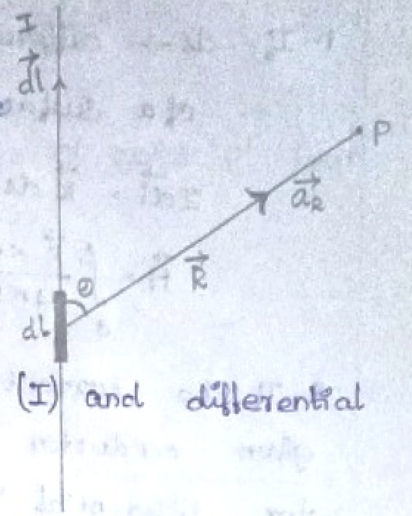
In general, Vector magnetic potential can be expressed as

$$\vec{A} = \frac{\mu}{4\pi} \int_V \left(\vec{J}/r\right) dv \; ; \; Wb/m.$$

BIOT-SAVART LAW:

It states that

the differential magnetic field $\vec{dH}$ of a conductor at point 'P' is



→ directly proportional to the product of current (I) and differential length (dl).

→ directly proportional to the sine of angle between the conductor and the line joining point 'P' and dl.

→ Inversely proportional to the square of the distance between point 'P' and dl.

$$\vec{dH} \propto \frac{I\, \vec{dl} \sin\theta}{R^2}$$

$$\vec{dH} = \frac{k\, I \vec{dl} \sin\theta}{R^2} \qquad \because k = 1/4\pi$$

$$\vec{dH} = \frac{I \vec{dl} \sin\theta}{4\pi R^2} \qquad \because \vec{dl} \times \vec{a_R} = dl \sin\theta.$$

$$\vec{dH} = \frac{I(\vec{dl} \times \vec{a_R})}{4\pi R^2} \qquad \because \vec{a_R} = \frac{\vec{R}}{|\vec{R}|} = \frac{\vec{R}}{R}.$$

$$= \frac{I(\vec{dl} \times \vec{R})}{4\pi R^3}.$$

$$\vec{H} = \int_l \frac{I(\vec{dl} \times \vec{R})}{4\pi R^3} \; A/m$$

Biot savart law in terms of distributed source

1. If $ds \rightarrow$ differential surface area considered of a surface having current density $\vec{K}$.

$$I\vec{dl} = \vec{K}\,ds.$$

$$\vec{H} = \int_S \frac{\vec{K}\times\vec{a_R}}{4\pi R^2}\,ds;\ A/m.$$

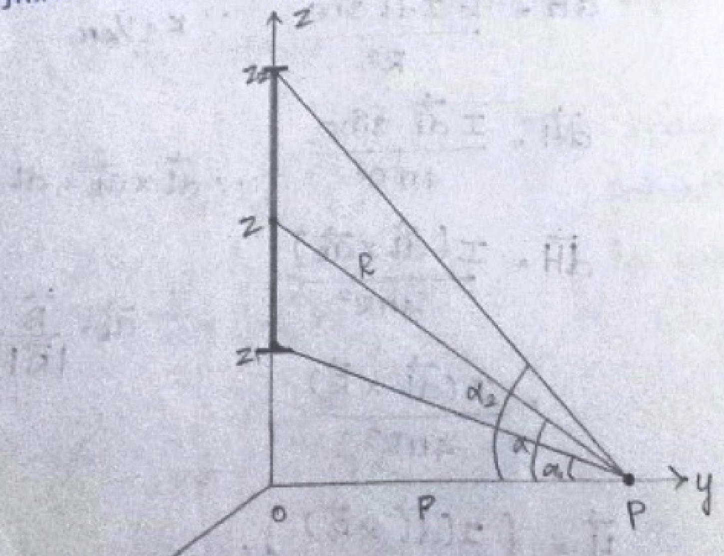
2. If the current density in a volume of a given conductor is $\vec{J}$ measured in A/m^2, then for differential Volume 'dv'

$$I\vec{dl} = \vec{J}\,dv$$

$$\vec{H} = \int_V \frac{\vec{J}\times\vec{a_R}}{4\pi R^2}\,dv\ ;A/m.$$

Applications of Biot-savart Law:

1. $\vec{H} \rightarrow$ Magnetic field intensity due to infinite and finite straight line conductor



Consider a conductor of finite length & placed along z-axis, Which carries current 'I'.

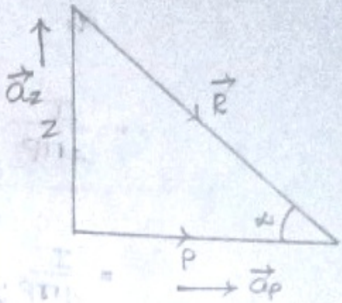
The conductor is placed such that its one end is at $z=z_1$ While other end at $z=z_2$

Consider a differential length $\vec{dl}$ along z-axis at a distance 'z' from origin

$$\vec{dl} = dz\,\vec{a}_z$$

Perpendicular distance of point 'P' from z-axis is 'R'

$$\vec{R} = \rho\vec{a}_\rho - z\vec{a}_z$$



$\tan\alpha = z/\rho$

$$R = |\vec{R}| = \sqrt{\rho^2 + z^2}$$

$$\vec{dl} \times \vec{R} = \begin{vmatrix} \vec{a}_\rho & \vec{a}_\phi & \vec{a}_z \\ 0 & 0 & dz \\ \rho & 0 & -z \end{vmatrix}$$

$$= \vec{a}_\phi(\rho dz).$$

$$\vec{dl} \times \vec{R} = \rho dz\,\vec{a}_\phi \longrightarrow (1)$$

By Biot Savart Law:

$$\vec{dH} = \frac{I\vec{dl} \times \vec{R}}{4\pi R^3} \longrightarrow (2)$$

Sub equ (1) in equ (2)

$$\vec{dH} = \frac{I\rho dz\,\vec{a}_\phi}{4\pi(\rho^2+z^2)^{3/2}}$$

$$\vec{H} = \int \frac{I\rho\, dz\,\vec{a}_\phi}{4\pi(\rho^2+z^2)^{3/2}} \longrightarrow (3)$$

We know that:

$$z/\rho = \tan\alpha$$

$$z = \rho\tan\alpha$$

$$dz = \rho\sec^2\alpha\, d\alpha.$$

Sub it in equ (3)

$$\vec{H} = \int_{\alpha_1}^{\alpha_2} \frac{I}{4\pi}\left[\frac{\rho^2 \sec^2\alpha \, d\alpha}{(\rho^2+\rho^2\tan^2\alpha)^{3/2}}\right]\vec{a_\phi}$$

$$= \int_{\alpha_1}^{\alpha_2} \frac{I}{4\pi}\left(\frac{\rho^2 \sec^2\alpha \, d\alpha}{\rho^3 \sec^3\alpha}\right)\vec{a_\phi}$$

$$= \frac{I}{4\pi\rho}\int_{\alpha_1}^{\alpha_2} \frac{1}{\sec\alpha}\, d\alpha \, \vec{a_\phi}$$

$$= \frac{I}{4\pi\rho}\int_{\alpha_1}^{\alpha_2} \cos\alpha \, d\alpha \, \vec{a_\phi}$$

$$= \frac{I}{4\pi\rho}(\sin\alpha)_{\alpha_1}^{\alpha_2}$$

$$\vec{H} = \frac{I}{4\pi\rho}\left[\sin\alpha_2 - \sin\alpha_1\right]\vec{a_\phi} \;\; A/m.$$

Also, $\vec{B} = \mu\vec{H}$.

$$\vec{B} = \frac{\mu I}{4\pi\rho}\left[\sin\alpha_2 - \sin\alpha_1\right]\vec{a_\phi}. \;\; Wb/m^2$$

For Infinite line:

$$\alpha_1 = -\pi/2 \;;\; \alpha_2 = \pi/2.$$

$$\vec{H} = \frac{I}{4\pi\rho}\left[\sin\pi/2 - \sin(-\pi/2)\right]\vec{a_\phi}$$

$$= \frac{2I}{4\pi\rho}\vec{a_\phi}$$

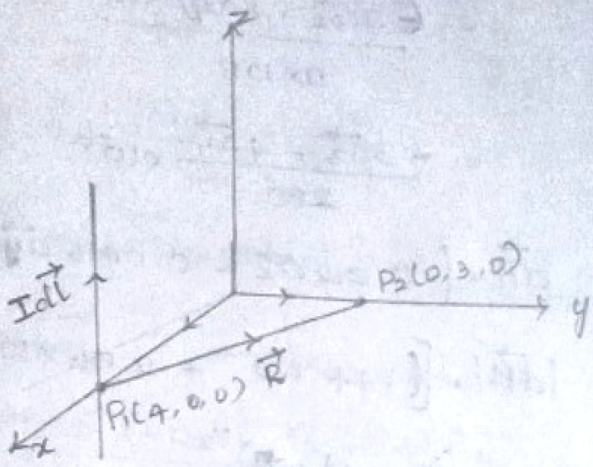
$$\vec{H} = \frac{I}{2\pi\rho}\vec{a_\phi} \;;\; A/m.$$

$$\vec{B} = \mu\vec{H}.$$

$$\vec{B} = \frac{\mu I}{2\pi\rho}\vec{a_\phi} \;;\; Wb/m^2.$$

Find the incremental $\vec{dH}$ field strength at point P_2 due to current element ($I\vec{dl}$) of $2\pi\vec{a_z}$ μA at 'P_1'. The coordinates of P_1 and P_2 are $(4,0,0)$ and $(0,3,0)$.

Soln:



$$\vec{P_1} = 4\vec{a_x} + 0\vec{a_y} + 0\vec{a_z}$$

$$\vec{P_2} = 0\vec{a_x} + 3\vec{a_y} + 0\vec{a_z}$$

$$dH \propto \frac{I\vec{dl}\sin\theta}{R^2}$$

$$\vec{dH} = \frac{I\vec{dl} \times \vec{R}}{4\pi R^3}$$

$$\vec{R} = \vec{R_{12}} = \vec{R_2} - \vec{R_1}$$

$$= \vec{P_2} - \vec{P_1} \qquad R = \sqrt{9+16}$$

$$= 3\vec{a_y} - 4\vec{a_x} \qquad R = 5$$

$$I\vec{dl} = 2\pi\vec{a_z}\ \mu A$$

$$I\vec{dl} \times \vec{R} = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ 0 & 0 & 2\pi\times10^{-6} \\ -4 & 3 & 0 \end{vmatrix}$$

$$= \vec{a_x}(-6\pi\times10^{-6}) + \vec{a_y}(-8\pi\times10^{-6}) + 0\vec{a_z}$$

$$= (-6\pi\vec{a_x} - 8\pi\vec{a_y})\times10^{-6}$$

$$d\vec{H} = \frac{(-6\pi\vec{a_x} - 8\pi\vec{a_y}) \times 10^{-6}}{4\pi \times 5\times5\times5}$$

$$= \frac{(-3\vec{a_x} - 4\vec{a_y}) \times 10^{-6}}{2\times 125}$$

$$= \frac{-3\vec{a_x} - 4\vec{a_y}}{250} \times 10^{-6} = (-12\vec{a_x} - 16\vec{a_y})\ nA/m$$

$$d\vec{H} = (-0.012\vec{a_x} - 0.016\vec{a_y}) \times 10^{-6}$$

$$|d\vec{H}| = [(1.44\times10^{-4} + 2.56\times10^{-4})\times10^{-12}]^{1/2}$$

$$= ((4\times10^{-4})10^{-12})^{1/2}$$

$$= 2\times10^{-8} = 20\,nA/m$$

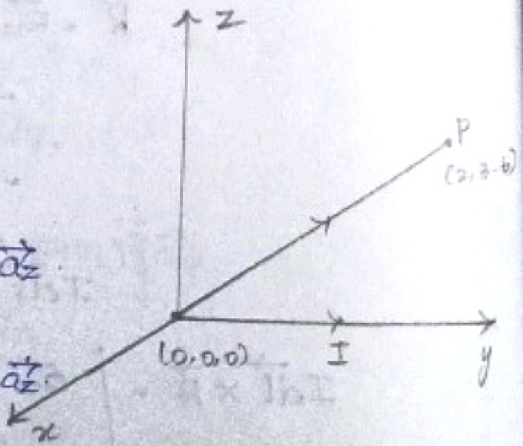
2. Calculate magnetic field intensity $\vec{H}$ at $(2,3,-6)$ due to current element of length 2m located at the origin in free space that carries current 10mA in the y direction.

Soln:

$$d\vec{H} = \frac{I\,d\vec{l} \times \vec{R}}{4\pi R^3}$$

$$\vec{R_{12}} = \vec{R_2} - \vec{R_1}$$

$$\vec{R_{12}} = 2\vec{a_x} + 3\vec{a_y} - 6\vec{a_z}$$



$$d\vec{l} = dx\,\vec{a_x} + dy\,\vec{a_y} + dz\,\vec{a_z}$$

$$d\vec{l} = 2\vec{a_y}$$

$$d\vec{l} \times \vec{R} = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ 0 & 2 & 0 \\ 2 & 3 & -6 \end{vmatrix}$$

$$= -12\vec{a_x} - \vec{a_y}(0) + \vec{a_z}(-4)$$

$$= -12\vec{a_x} - 4\vec{a_z}$$

$$|\vec{R}| = \sqrt{4+9+36}$$
$$= 7.$$

$$d\vec{H} = \frac{10(-12\vec{a_x} - 4\vec{a_z})}{4\pi \times 7^3}.$$

$$= \frac{-120\vec{a_x} - 40\vec{a_z}}{4308.08}.$$

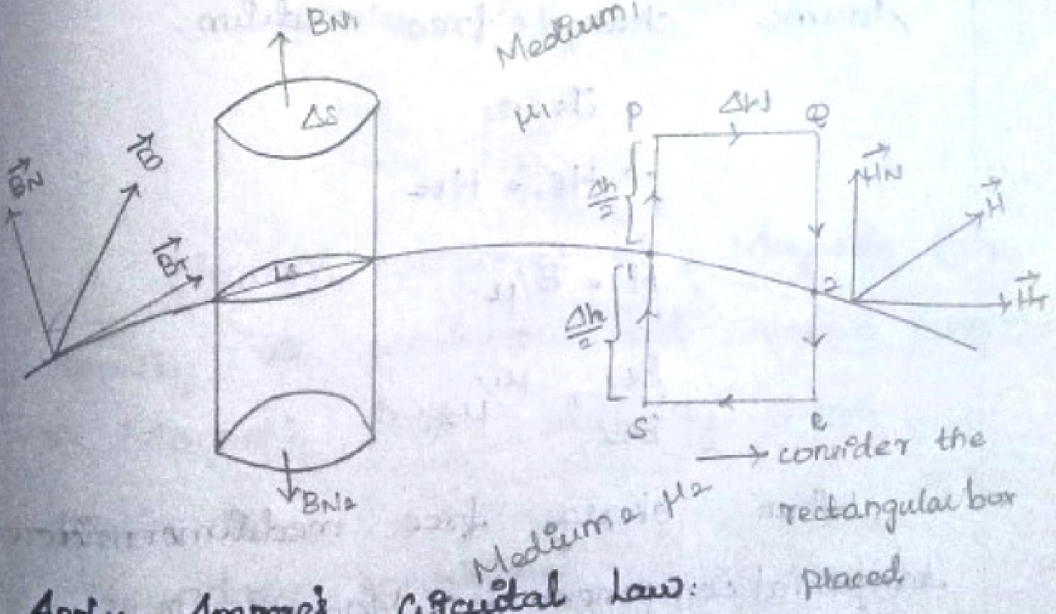
$$\vec{H} = -0.02785\vec{a_x} - 0.0092848\,\vec{a_z}\ A/m.$$

$$\vec{B} = \mu_0 \vec{H}$$ *free space ; $\mu_r = 1$

$$\vec{B} = 4\pi \times 10^{-7}(-0.02\,\vec{a_x} - 0.01\,\vec{a_z})$$

Boundary condition of Magnetic field :

The condition exiting at boundary of both medium when field passes from one medium to another medium.



→ consider the rectangular box placed symmetrically in both Medium

Apply Ampere's circuital Law:

$$\oint \vec{H} \cdot d\vec{l} = I.$$

We can split it as follows:

$$I = J_s \Delta w.$$

$$\oint_P \vec{H}\cdot\vec{dl} = \int_P^Q \vec{H}\cdot\vec{dl} + \int_Q^1 \vec{H}\cdot\vec{dl} + \int_1^R \vec{H}\cdot\vec{dl} + \int_R^S \vec{H}\cdot\vec{dl} +$$

$$\int_S^2 \vec{H}\cdot\vec{dl} + \int_2^P \vec{H}\cdot\vec{dl} = I = J_s\Delta w.$$

$$\oint_P \vec{H}\cdot\vec{dl} = H_{t_1}\oint dl - H_{N_1}\oint dl - H_{N_2}\oint dl - H_{t_2}\oint dl.$$

$$+ H_{N_2}\oint dl + H_{N_1}\oint dl$$

$$= H_{t_1}\Delta w - H_{N_1}\frac{\Delta h}{2} - H_{N_2}\frac{\Delta h}{2} - H_{t_2}\Delta w$$

$$+ H_{N_2}\frac{\Delta h}{2} + H_{N_1}\frac{\Delta h}{2} = J_s\Delta w$$

$$\Rightarrow H_{t_1}\Delta w - H_{t_2}\Delta w = J_s\Delta w$$

$$H_{t_1} - H_{t_2} = J_s$$

In general, the tangential component of magnetic field intensity is not continuous.

Assume charge free medium.

$$J_s = 0$$

$$\therefore H_{t_1} = H_{t_2}$$

$$\because H = B/\mu.$$

$$\frac{B_{t_1}}{B_{t_2}} = \mu_1/\mu_2.$$

When charge free medium considered, tangential component of Magnetic field intensity is continuous but tangential component of magnetic flux density is not continuous.

Apply Gauss law of Magnetostatics

$$\oint \vec{B} \cdot \vec{ds} = 0$$

We can split it as

$$\oint_{top} \vec{B} \cdot \vec{ds} + \int_{bottom} \vec{B} \cdot \vec{ds} + \oint_{LS} \vec{B} \cdot \vec{ds} = 0.$$

$$\Delta h \to 0$$

$$\oint_{LS} \vec{B} \, \vec{ds} = 0$$

$$\oint_{top} \vec{B} \cdot \vec{ds} + \oint_{bottom} \vec{B} \cdot \vec{ds} = 0.$$

$$B_{N1} (\Delta s) - B_{N2} (\Delta s) = 0$$

$$B_{N1} = B_{N2}.$$

$$\because B = \mu H$$

$$\mu_1 H_{N1} = \mu_2 H_{N2}.$$

$$\frac{H_{N1}}{H_{N2}} = \frac{\mu_2}{\mu_1}$$

$\therefore$ Normal component of Magnetic flux density is continuous but Normal component of Magnetic field intensity is not continuous.

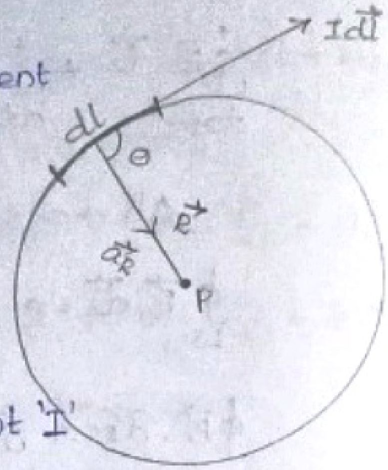
Boundary conditions:

1). $H_{t1} = H_{t2}$ 3) $B_{N1} = B_{N2}$.

2). $\frac{B_{t1}}{B_{t2}} = \frac{\mu_1}{\mu_2}$ 4) $\frac{H_{N1}}{H_{N2}} = \mu_2/\mu_1$

Applications Biot-savart law:

2. $\vec{H}$ (Magnetic field intensity) at the center of a circular conductor:



Consider a current carrying conductor arranged in a circular form.

The conductor carries a direct current 'I'

$\vec{dl} \Rightarrow$ differential length.

We need to obtain $\vec{H}$ at a point 'P'.

$\theta \Rightarrow$ angle between $I\vec{dl}$ and $\vec{R}$

$\theta = 90°$

$\vec{R} \rightarrow$ distance Vector.

By Biot-savart Law:

$$d\vec{H} = \frac{I\vec{dl} \times \vec{R}}{4\pi R^3}$$

$$d\vec{H} = \frac{I\vec{dl} \times \vec{a_R}}{4\pi R^2}$$

$$d\vec{H} = \frac{I dl \sin\theta}{4\pi R^2} \vec{a_N}$$

$$\vec{H} = \oint \frac{I dl \sin\theta}{4\pi R^2} \vec{a_N}$$

$$\vec{H} = \frac{I \sin\theta}{4\pi R^2} \vec{a_N} \oint dl$$

$$= \frac{I \sin\theta}{4\pi R^2} \vec{a_N} (2\pi R).$$

$$\vec{H} = \frac{I\sin\theta}{2R}\vec{a_N}.$$

Here, $\theta = 90°$

$$\vec{H} = \frac{I}{2R}\vec{a_N}.$$

Note:

If the circular loop is placed in xy plane, $\vec{a_N} = \vec{a_z}$.

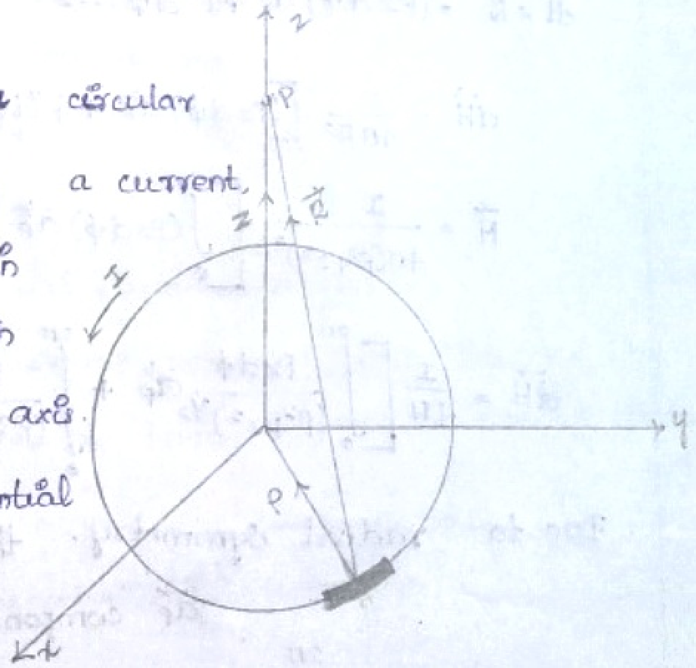
$$\vec{H} = \frac{I}{2R}\vec{a_z} \; ; A/m.$$

$$\vec{B} = \mu\vec{H}$$

$$\vec{B} = \frac{\mu I}{2R}\vec{a_z} \; ; Wb/m^2$$

---

3. Magnetic field intensity, $\vec{H}$, on the axis of a circular loop:



→ Consider a circular loop carrying a current 'I' placed in xy plane with z-axis as the axis.

→ $\vec{dl}$ → differential length on the circular loop

→ ρ → radius of the circular loop

→ P → point located on z-axis from the origin at a distance 'z'.

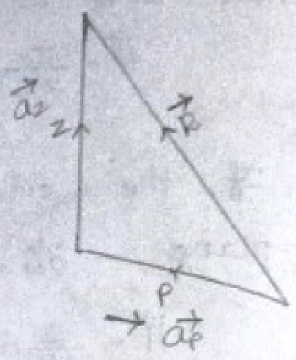
We have to find $\vec{H}$ at point 'P';

By Biot-Savart Law,

$$\vec{dH} = \frac{I\vec{dl} \times \vec{a_R}}{4\pi R^2}$$

$$d\vec{H} = \frac{I\,\vec{dl} \times \vec{R}}{4\pi R^3}$$

$$\therefore \vec{dl} = \rho\, d\phi\, \vec{a_\phi}$$



$$\vec{R} = -\rho\vec{a_\rho} + z\vec{a_z}$$

$$|\vec{R}| = R = \sqrt{(\rho^2 + z^2)}$$

$$\vec{dl} \times \vec{R} = \begin{vmatrix} \vec{a_\rho} & \vec{a_\phi} & \vec{a_z} \\ 0 & \rho d\phi & 0 \\ -\rho & 0 & z \end{vmatrix}$$

$$= \vec{a_\rho}(\rho z d\phi) + \vec{a_z}(\rho^2 d\phi).$$

$$\vec{dl} \times \vec{R} = (\rho z d\phi)\vec{a_\rho} + (\rho^2 d\phi)\vec{a_z}.$$

$$d\vec{H} = \frac{I}{4\pi R^3}\left[(\rho z d\phi)\vec{a_\rho} + (\rho^2 d\phi)\vec{a_z}\right].$$

$$\vec{H} = \frac{I}{4\pi(\rho^2+z^2)^{3/2}}\left[\int_0^{2\pi}(\rho z d\phi)\vec{a_\rho} + \int_0^{2\pi}(\rho^2 d\phi)\vec{a_z}\right]$$

$$d\vec{H} = \frac{I}{4\pi}\left[\int_0^{2\pi}\frac{\rho z d\phi}{(\rho^2+z^2)^{3/2}}\vec{a_\rho} + \int_0^{2\pi}\frac{\rho^2 d\phi}{(\rho^2+z^2)^{3/2}}\vec{a_z}\right]$$

Due to radial symmetry, there is no $\vec{a_\rho}$ component.

$$\vec{H} = \frac{I}{4\pi}\int_0^{2\pi}\frac{\rho^2 d\phi}{(\rho^2+z^2)^{3/2}}\vec{a_z}$$

$$= \frac{I}{4\pi}\frac{\rho^2}{(\rho^2+z^2)^{3/2}}\int_0^{2\pi} d\phi\ \vec{a_z}$$

$$= \frac{I}{4\pi}\frac{\rho^2}{(\rho^2+z^2)^{3/2}}(\phi)_0^{2\pi}\ \vec{a_z}$$

$$\vec{H} = \frac{I}{2}\frac{\rho^2}{(\rho^2+z^2)^{3/2}}\vec{a_z}\ ;\ A/m$$

$$\vec{B} = \mu\vec{H}$$

$$\vec{B} = \frac{\mu I}{2}\frac{\rho^2}{(\rho^2+z^2)^{3/2}}\vec{a_z} \text{ ; Wb/m}^2$$

Inductance:

→ The element which can be stored the energy in terms of magnetic field is called Inductance.

→ The element which can be stored the energy interms of electric field is called capacitance.

→ Two types of Inductance

Self Inductance.

Mutual Inductance

Self Inductance:

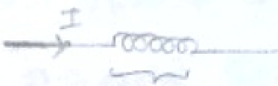
$\lambda = N\phi$

λ = total flux linkage

ϕ - flux across each turn (the coil by current flowing through it)

$N \to$ No. of turns.

Lenz Law:



$$e_{induced} = -\frac{d\phi\lambda}{dt}.$$

$$= -\frac{dN\phi}{dt}$$

$$e_{ind} = -N\frac{d\phi}{dt}.$$

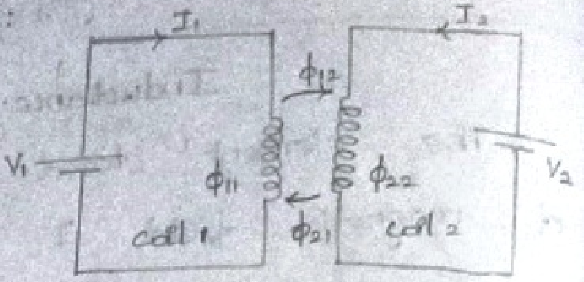
$$= -N\frac{d\phi}{dI}\cdot\frac{dI}{dt}$$

$$L = N\frac{d\phi}{dI}$$

$$e_{ind} = -L\frac{dI}{dt}$$ ∴ $L \to$ self Inductance.

$\therefore L = \frac{N\phi}{I}$; Henry.

Mutual Inductance:



$\phi_{11} \to$ flux across coil 1 due to I_1

$\phi_2 \to$ flux across coil 2 due to I_2.

$\phi_{12} \to$ flux linked with coil 2 due to I_1

$\phi_{21} \to$ flux linked with coil 1 due to I_2.

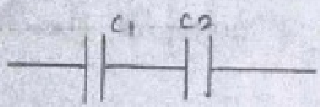
$M_{12} \to$ Mutual Inductance of coil 2.

$M_{21} \to$ Mutual Inductance of coil 1.

$$M_{12} = M_2 \frac{\phi_{12}}{I_1} \text{ ; Henry}$$

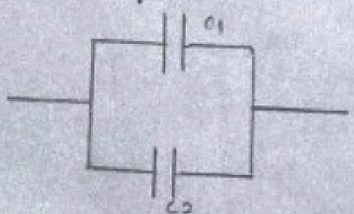
$$M_{21} = M_1 \frac{\phi_{21}}{I_2} \text{ ; Henry.}$$

Two capacitance in series:



$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

Two capacitance in Parallel:

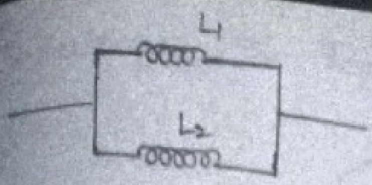


$$C_{eq} = C_1 + C_2.$$

Two Inductance in series:

$$L_{eq} = L_1 + L_2.$$

Two inductance in parallel:



$$L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$$

Two inductance in series with +ve coupling: (series Aiding) 

$$L_{eq} = L_1 + L_2 + 2M.$$

(or) 

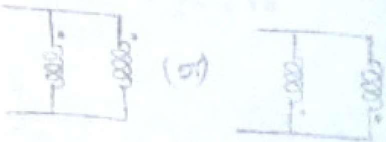
Two Inductance in series with -ve coupling: (series opposing) 

$$L_{eq} = L_1 + L_2 - 2M.$$

(or) 

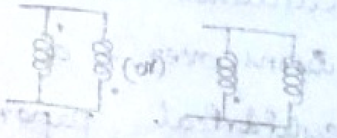
Two Inductance in parallel with +ve coupling: (parallel Aiding)

$$L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M}$$



Two Inductance in parallel with -ve coupling: (parallel opposing)

$$L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 + 2M}$$



1. 2 coils with negligible resistance and self Inductances are 0.2 and 0.1 H respect. are connected in series and parallel. If their M.I. is 0.1 H. Determine the effective Inductance of their combination in each case.

$L_1 = 0.2H$ $\quad M = 0.1H$

$L_2 = 0.1H.$

1). $L_{eq} = L_1 + L_2$

$= 0.3H$

2). $L_{eq} = \frac{L_1 L_2}{L_1 + L_2} = \frac{0.02}{0.3}$

$= 0.066H$

3) $L_{eq} = L_1 + L_2 + 2M$

$= 0.3 + 0.2$

$= 0.5H$

4) $L_{eq} = L_1 + L_2 - 2M$

$= 0.1H$

5) $L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M}$

$= \frac{0.02 - 0.01}{0.1}$

$= 0.1H$

6) $L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 + 2M}$

$= \frac{0.01}{0.5}$

$= 0.02H$

1) solenoid.
2) Toroid
3) coaxial cable
4) Transmission lines

$\vec{H}$ & $\vec{B}$ expression and deduce its expression for Inductance

determine Inductance

1 Calculate the Inductance of solenoid of 2500 turns over a length of 0.5m and a cylindrical paper tube with the 4cm in the diameter. Assume the medium is air.

Soln:

$l = 0.5m$.

$r = 2 \times 10^{-2} m$.

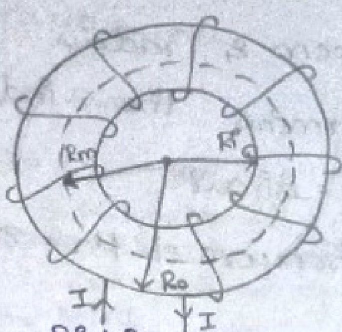
$N = 2500$

$L = \frac{N^2 \mu A}{l}$

$= \frac{2500 \times 2500 \times 4\pi \times 10^{-7} \times \pi \times 4 \times 10^{-4}}{0.5}$

$= \frac{25 \times 25 \times 16 \times \pi^2 \times 10^{-6}}{5} = 19.7mH$

Inductance of Toroid:



Consider a toroid of 'N' No. of turns carrying current 'I' with radius 'R'

$L = N\phi/I$

$$R_m = \frac{R_i + R_o}{2} \rightarrow \text{Mean Radius.}$$

$$\oint_l \vec{H} \cdot \vec{dl} = I \rightarrow \text{From Ampere's circuital Law.}$$

$$H = NI/_{2\pi R} \; ; A/m.$$

Where $l \rightarrow$ length of coil

$$B = \mu H = \frac{\mu NI}{2\pi R} \quad l = 2\pi R$$

The Magnetic flux is given as

$$\phi = BA = \frac{\mu NI}{2\pi R} \cdot A$$

Where $B \rightarrow$ flux density

We know that

$$L = \frac{N\phi}{I} = \frac{N^2 \mu A}{2\pi R}$$

$A \rightarrow$ cross sectional Area

if $A = \pi r^2$

$r \rightarrow$ radius of the coil

$$L = \frac{N^2 \mu (\pi r^2)}{2\pi R}$$

$$L = \frac{N^2 \mu r^2}{2R}$$

Where.

$$L = \frac{\mu N^2 r^2}{2R} \; ; \text{Henry.}$$

$r \rightarrow$ radius of coil
$R \rightarrow$ radius of Toroid.
$N \rightarrow$ Number of turns.

Definition:

It is a hollow circular ring on which a large no. of turns of wire are closely wound. A toroid is a coil wound in the shape of torous. It is used as an inductor in electric circuits where large inductances are needed.

2. A Toroid of 1000 turns has mean radius of 20cm & radius for the winding of 2cm determine the inductance when the medium is air:

2). Iron core of $\mu_r = 800$.

Soln:

$N = 1000$

$R_m = 20cm = 20\times10^{-2} m.$

$r = 2\times10^{-2} m.$

1). $L = \frac{\mu N^2 r^2}{2R}$

$= \frac{4\pi\times10^{-7}\times10^{6}\times4\times10^{-4}}{2\times20\times10^{-2}}$

$= 12.56 \times10^{-4}$

$L = 1.256 mH$

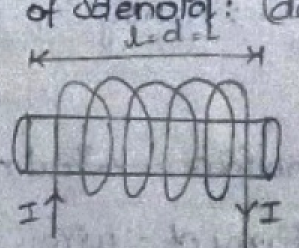
2). $L = \frac{10^{6}\times800\times4\pi\times10^{-7}\times4\times10^{-4}}{40\times10^{-2}}$

$= 100.48\times10^{9}\times10^{-11}$

$= 100.48\times10^{-2}$

$L = 1H.$

Inductance of Solenoid: (defn).



Consider a solenoid of 'N' number of turns carrying the current 'I'

From Ampere's circuital law,

$$\oint_l \vec{H}\cdot\vec{dl} = I.$$

$$Hl = NI$$

$$H = NI/l \longrightarrow ④$$

We know that:

$$\phi = BA \longrightarrow ②$$

Magnetic flux density

Where, $B = \mu H \longrightarrow ③$

Sub equ ④ in equ ③

$$B = \frac{\mu NI}{l} \longrightarrow ⑤$$

Equ ② becomes

$$\therefore \phi = \frac{\mu NIA}{l}$$

Self Inductance

$$L = N\phi / I \longrightarrow ①$$

$$= \frac{N^2 \mu IA}{l \cdot I}$$

$\phi \rightarrow$ flux linkage through solenoid

$$L = \frac{\mu N^2 A}{l} \text{ : Henry}$$

Where,

Definition:

(It is a current carrying coil of wire in cylindrical form.

It acts like magnet when a current passes through it.)

Where, $N \rightarrow$ Number of turns

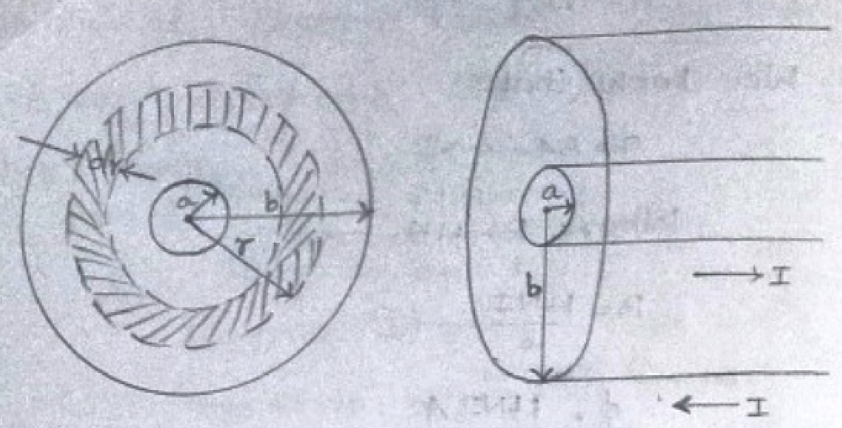
$A \rightarrow$ Cross sectional Area of solenoid.

$l \rightarrow$ length of solenoid.

$$L = \frac{\mu N^2 \pi r^2}{l}$$

$r \rightarrow$ radius of coil.

Inductance of Coaxial Cable:



Consider a coaxiable cable of inner radius 'a' carries a current of 'I' and outer radius 'b'. carries a current of '-I'.

Consider a annular ring of thickness dr at a distance r from inner conductor.

Self inductance is defined as

$$L = N\phi / I \longrightarrow ①$$

For a coaxial cable, N=1

$$L = \phi / I \longrightarrow ②$$

The total Magnetic flux linkage is given as.

$$\phi = \int B . ds \longrightarrow ③$$

Where,
B → Magnetic flux density

$$B = \mu H \longrightarrow ④$$

H → Magnetic field Intensity.

By ampere's circuital law,

$$\int \vec{H} . \vec{dl} = I.$$

$$H = NI / l \longrightarrow ⑤$$

Equ ④ becomes

$$B = \frac{\mu N I}{l}$$

We know : $N = 1$

$l = 2\pi r$

$$B = \frac{\mu I}{2\pi r} \rightarrow ⑥$$

Equ ③ becomes

$$\phi = \int_a^b \frac{\mu I}{2\pi r} dr$$

$$= \frac{\mu I}{2\pi} \int_a^b \frac{1}{r} dr$$

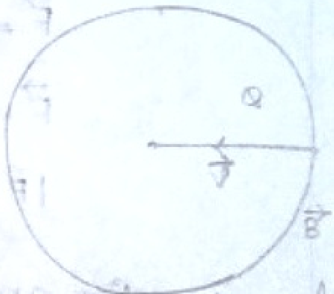
$$\phi = \frac{\mu I}{2\pi} \ln(b/a) \longrightarrow ⑦$$

Sub. equ ⑦ in equ ②

$$① \Rightarrow L = \frac{\mu}{2\pi} \ln(b/a); \text{ H/m}$$

Magnetic force:

$$\vec{F_m} = Q[\vec{V} \times \vec{B}]$$



The Magnetic force can be defined as the attractive or repulsive force that is exerted b/w the poles of a magnet and electrically charged moving particles.

Force on differential current element:

$$d\vec{F} = dQ(\vec{V} \times \vec{B})$$

$dQ = \rho_v dv \rightarrow$ charge in terms of Volume charge density.

$$d\vec{F} = \rho_v dv (\vec{V} \times \vec{B})$$

$$d\vec{F} = \rho_v (\vec{V} \times \vec{B}) dv \qquad \because \vec{J} = \rho_v \vec{V}$$

$$d\vec{F} = (\vec{J} \times \vec{B})\, dv$$

$$\vec{F} = \int_V (\vec{J} \times \vec{B})\, dv.$$

↳ $\vec{F}$ in terms of Volume of charge density.

In terms of surface charge density:

$$\vec{F} = \int_S (\vec{K} \times \vec{B})\, ds.$$

In terms of line charge density:

$$\because \vec{J}\, dv = I\, d\vec{l}$$

$$\vec{F} = \oint_l I (d\vec{l} \times \vec{B}).$$

For uniform Magnetic field:

$$d\vec{l} = \vec{L}$$

$$\vec{B} = \vec{B}$$

$$\vec{F} = I(\vec{L} \times \vec{B}).$$

$$\vec{F} = IBL \sin\theta\, \hat{n}$$

$$|\vec{F}| = F = ILB \sin\theta.$$

$\theta \Rightarrow$ angle b/w $\underbrace{\text{magnetic field}}_{\text{conductor}}$ and line joining point and conductor.

$L \Rightarrow$ Length of Conductor.

$I \rightarrow$ Current flowing through the conductor.

1. A conductor of 6m long lies along z direction with a current of 2A in $\vec{a_z}$. find the force experienced by the conductor if

$\vec{B} = 0.08\vec{a_x}$ Tesla

Soln:

$\vec{B} = 0.08\vec{a_x}$ T

$L = 6\vec{a_z}$ m

$I = 2A.$

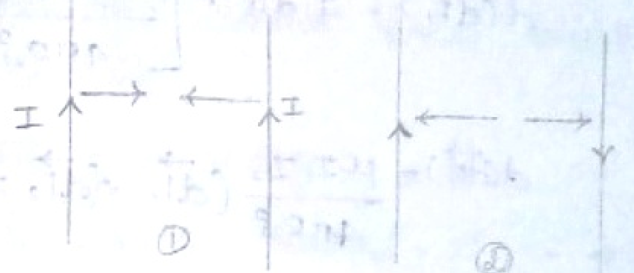
$$\vec{L} \times \vec{B} = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ 0 & 0 & 6 \\ 0.08 & 0 & 0 \end{vmatrix} = +0.48\vec{a_y}$$

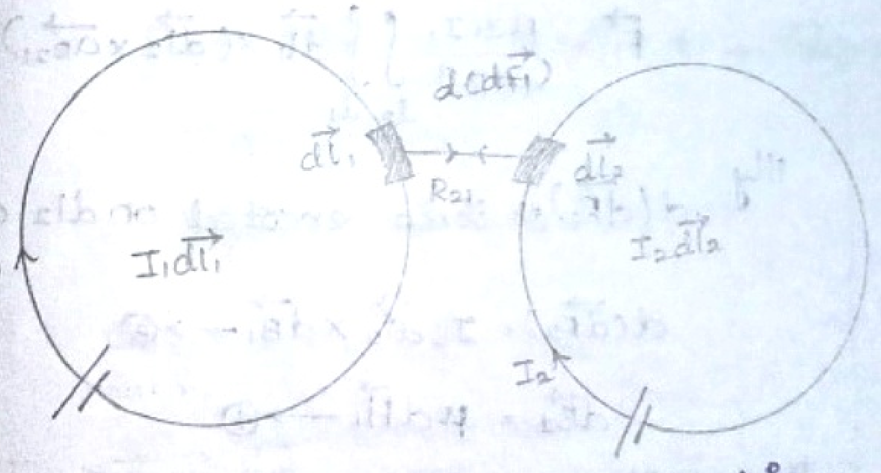
$\vec{F} = I(\vec{L} \times \vec{B})$

$= 2(0.48\vec{a_y})$

$\vec{F} = 0.96\vec{a_y}$ N.

Magnetic force between parallel lines:





$dB_2 \rightarrow$ Magnetic flux density of line 2 due to current I_2.

$dH_2 \rightarrow$ Magnetic field intensity of line 2 due to current I_2.

$dB_1 \rightarrow$ Magnetic flux density of line 1 due to current I_1.

$dH_1 \rightarrow$ Magnetic field intensity of line 1 due to current I_1.

$d(d\vec{F_1}) \rightarrow$ Force exerted on dl_1 due to dB_2

$$d(d\vec{F_1}) = I_1 d\vec{l_1} \times d\vec{B_2} \longrightarrow (1)$$

$$d\vec{B_2} = \mu d\vec{H_2} \longrightarrow (2)$$

By Biot Savart Law,

$$d\vec{H_2} = \frac{I_2 d\vec{l_2} \times \vec{a}_{R21}}{4\pi R_{21}^2}$$

$$d\vec{B_2} = \frac{\mu I_2 d\vec{l_2} \times \vec{a}_{R21}}{4\pi R_{21}^2}$$

$$d(d\vec{F_1}) = I_1 d\vec{l_1} \times \left[\frac{\mu I_2 d\vec{l_2} \times \vec{a}_{R21}}{4\pi R_{21}^2}\right]$$

$$d(d\vec{F_1}) = \frac{\mu I_1 I_2}{4\pi R_{21}^2}\left(d\vec{l_1} \times (d\vec{l_2} \times \vec{a}_{R21})\right)$$

$$\vec{F_1} = \frac{\mu I_1 I_2}{4\pi R_{21}^2} \int_{l_2}\int_{l_1} d\vec{l_1} \times (d\vec{l_2} \times \vec{a}_{R21}) \ : N$$

$||^{ly}$ $d(d\vec{F_2}) \rightarrow$ Force exerted on dl_2 due to dB_1.

$$d(d\vec{F_2}) = I_2 d\vec{l_2} \times d\vec{B_1} \longrightarrow (3)$$

$$d\vec{B_1} = \mu d\vec{H_1} \longrightarrow (4)$$

By Biot Savart Law.

$$d\vec{H_1} = \frac{I_1 d\vec{l_1} \times \vec{a}_{R12}}{4\pi R_{12}^2}$$

$$d\vec{B_2} = \frac{\mu I_1 d\vec{l_1} \times \vec{a}_{R12}}{4\pi R_{12}^2}$$

$$d(d\vec{F_2}) = I_2 d\vec{l_2} \times \left[\frac{\mu I_1 d\vec{l_1} \times \vec{a}_{R12}}{4\pi \vec{R_{12}}^2}\right]$$

$$d(d\vec{F_2}) = \frac{\mu I_1 I_2}{4\pi R_{12}^2}\left[d\vec{l_2} \times (d\vec{l_1} \times \vec{a_{R_{12}}})\right]$$

$$\vec{F_2} = \frac{\mu I_1 I_2}{4\pi R_{12}^2} \int_{l_2}\int_{l_1} d\vec{l_2} \times (d\vec{l_1} \times \vec{a_{R_{12}}}) \; ; \text{N}.$$

A current element $I_1 d\vec{l_1} = 10^{5}\vec{a_z}$ Am is located at $P_1(1,0,0)$. While a second element $I_2 d\vec{l_2} = 10^{5}(0.6\vec{a_x} - 2\vec{a_y} + 3\vec{a_z})$ Am is located at $P_2(-1,0,0)$. Assume that both are in free space find the vector force exerted on $I_2 d\vec{l_2}$ and $I_1 d\vec{l_1}$.

Soln: $P_1(1,0,0)$ and $P_2(-1,0,0)$

$$\vec{R_{21}} = +2\vec{a_x} \Rightarrow R_{21}^2 = 2^2 = 4 \Rightarrow |\vec{R_{21}}| = 2$$

$$\vec{R_{12}} = -2\vec{a_x} \Rightarrow R_{12}^2 = 2^2 = 4 \Rightarrow |\vec{R_{12}}| = 2$$

$$I_1 d\vec{l_1} = 10^{-5}\vec{a_z}$$

$$I_2 d\vec{l_2} = (0.6\vec{a_x} - 2\vec{a_y} + 3\vec{a_z})10^{-5}$$

$$\vec{a_{R_{21}}} = \frac{\vec{R_{21}}}{|\vec{R_{21}}|} = \frac{2\vec{a_x}}{2} = \vec{a_x}.$$

$$\vec{a_{R_{12}}} = \frac{\vec{R_{12}}}{|\vec{R_{12}}|} = \frac{-2\vec{a_x}}{2} = -\vec{a_x}.$$

1. $$d(d\vec{F_1}) = \frac{4\pi\times10^{-7}}{4\pi\times4}\left(10^{-5}\vec{a_z} \times \left[(0.6\vec{a_x} - 2\vec{a_y} + 3\vec{a_z})10^{-5} \times \vec{a_x}\right]\right.$$

$$I_2 d\vec{l_2} \times \vec{a_{R_{21}}} = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ 0.6 & -2 & 3 \\ 1 & 0 & 0 \end{vmatrix} = 3\vec{a_y} + 2\vec{a_z}$$

$$\vec{I_1 dl_1} \times (\vec{I_2 dl_2} \times \vec{a_{R_{21}}}) = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ 0 & 0 & 1 \\ 0 & 3 & 2 \end{vmatrix} = -3\vec{a_x}$$

$$\vec{F_1} = \frac{10^{-7}}{4} \times 10^{-10} (-3\vec{a_x}).$$

$$\vec{F_1} = -0.75 \times 10^{-17} \vec{a_x} \ N.$$

2. $$\vec{I_1 dl_1} \times \vec{a_{R12}} \Rightarrow \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{vmatrix}$$

$$= -\vec{a_y}$$

$$\vec{I_2 dl_2} \times (\vec{I_1 dl_1} \times \vec{a_{R12}}) = \begin{vmatrix} \vec{a_x} & \vec{a_y} & \vec{a_z} \\ 0.6 & -2 & 3 \\ 0 & -1 & 0 \end{vmatrix}$$

$$= 3\vec{a_x} - 0.6\vec{a_z}$$

$$\vec{F_2} = \frac{4\pi \times 10^{-7}}{4\pi \times 4} \times 10^{-10} (3\vec{a_x} - 0.6\vec{a_z})$$

$$\vec{F_2} = (0.75\vec{a_x} - 0.15\vec{a_z}) \times 10^{-17} \ N.$$

Magnetic Energy stored in Inductor:

$$\text{Power, } P = \frac{\text{Work}}{\text{time}}.$$

$$W = P \times t$$

By Work energy theorem,

$$E = Pt.$$

We know that:

$$P = VI.$$

$$E = V_L i t.$$

$$\because V_L = L\,di/dt..$$

$$E = L\frac{di}{dt}\, i t.$$

Differentiating with respect to 't'

$$dE = L\frac{di}{dt}\, i\, dt.$$

$$dE = L\, i\, di.$$

Integrating on both sides.

$$E = L\int_0^I i\, di$$

General expression of Energy stored in Inductor:

$$E = \frac{1}{2} L I^2.$$

The work required to move a unit magnet around the magnetic circuit is called magnetomotive force.

$$MMF = \underbrace{NI = Hl}_{\text{by Ampere's circuital Law}}$$

Inductance is the property of Inductor which oppose sudden flow of current.

$$L = \frac{N\phi}{I}.$$

$$\because E = \frac{1}{2}LI^2 = \frac{1}{2}\frac{N\phi}{I}I^2$$

$$= \frac{1}{2}N\phi I.$$

$$= \frac{1}{2}Hl\phi.$$

$$\because \phi = BA.$$

$$E = \frac{1}{2}BAHl.$$

Energy per unit Volume $= \frac{E}{Al} = \frac{1}{2} BH$

$\because B = \mu H$.

$$\frac{E}{Al} = \frac{1}{2}\mu H^2 = \frac{1}{2} B^2/\mu.$$

An air-cored solenoid having ~~dielectric~~ diameter of 4cm and length of 60cm is wound with 4000 turns if a current of 5A flows in the solenoid. calculate the Inductance and Energy stored in an inductor.

Soln:. $l = 60cm$ $\quad l = 60 \times 10^{-2}$ m.

$N = 4000$ turns.

$I = 5A$.

$d = 4cm$. $\quad r = 2 \times 10^{-2}$ m.

$A = \pi r^2$

$= 3.14 \times 4 \times 10^{-4}$

$A = 12.56 \times 10^{-4}$ m^2.

$NI = Hl$.

$4000 \times 5 = H \times 60 \times 10^{-2}$.

$$H = \frac{20 \times 10^3}{60 \times 10^{-2}}$$

$= 0.33 \times 10^5$.

$\phi = BA$.

$= \mu \times H \times A$.

$= 4\pi \times 10^{-7} \times 0.33 \times 10^5 \times 12.56 \times 10^{-4}$.

$\phi = 52.058 \times 10^{-6}$

$$L = \frac{N\phi}{I} = \frac{4000 \times 52.058}{5} \times 10^{-6}.$$

$L = 41.6464 \, mH$

$E = \frac{1}{2} L I^2$

$= \frac{1}{2} \times 41.6464 \times 25 \times 10^{-3}$

$= 520.58 \times 10^{-3} J$

$E = 0.5 J = 0.52 J$

Torque:

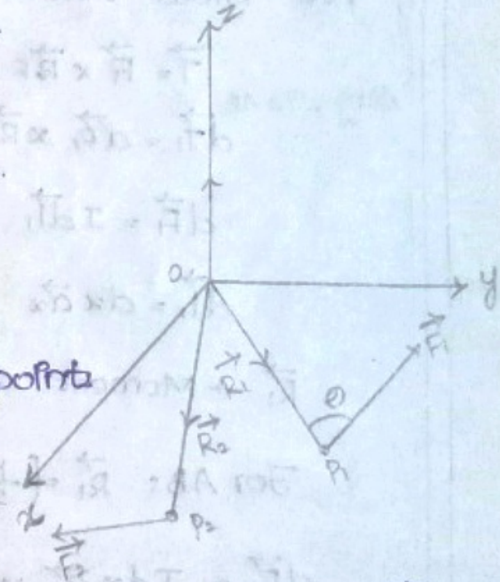
Torque is directly proportional to the force, perpendicular distance and sine angle between $\vec{F}$ and $\vec{R}$.

$T \propto FR \sin\theta$

$T = FR \sin\theta$

$\vec{T} = FR \sin\theta \, \hat{n}$

$\vec{T} = \vec{R} \times \vec{F}$



To find Net Torque:

Consider two points located at P_1 and P_2.

To find Net Torque due to $\vec{F_1}$ and $\vec{F_2}$:

Consider : $\vec{F_1} = -\vec{F_2}$

Consider $\vec{F_1}$ and $\vec{F_2}$ have same magnitude but opposite direction, then $\vec{F_1}$ and $\vec{F_2}$ forms a dipole.

$\vec{T} = (\vec{R_1} \times \vec{F_1}) + (\vec{R_2} \times \vec{F_2})$

$= (\vec{R_1} \times \vec{F_1}) + (\vec{R_2} \times (-\vec{F_1}))$

$$\vec{T} = (\vec{R_1} - \vec{R_2}) \times \vec{F_1}$$

$$\vec{T} = \vec{R_{21}} \times \vec{F_1}$$

Torque definition:

When a current loop is placed parallel to a magnetic field, force acts on the loop that tend to rotate it.

The tangential force multiplied by the radial distance at which it acts is called Torque or Mechanical moments on the loop.

The moment of force or torque about a specified point is defined as Vector product of moment arm $\vec{R}$ and force $\vec{F}$.

$$\vec{T} = \vec{R} \times \vec{F}.$$

Magnetic Moment:

Magnetic moment is defined as the maximum torque per magnetic flux density.

$$m = T/B$$

Determine the expression of torque on a differential current loop in a uniform magnetic field.

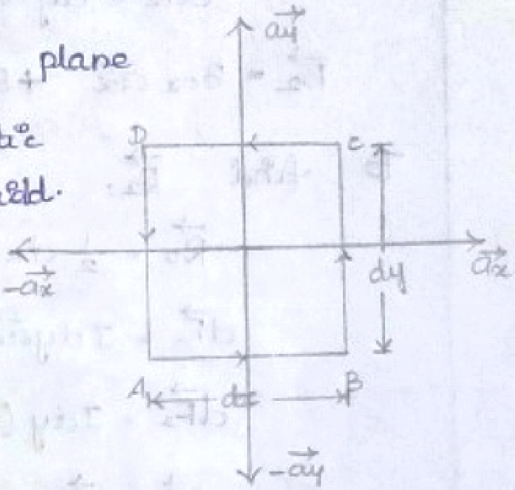
$d\vec{T_1} \rightarrow$ differential torque along side AB

$d\vec{T_2} \rightarrow$ differential torque along side BC

$d\vec{T_3} \rightarrow$ differential torque along side CD.

$d\vec{T_4} \rightarrow$ differential torque along side DA

Lets us consider a differential current loop in xy plane with a uniform magnetic field.



$d\vec{T} = d\vec{T_1} + d\vec{T_2} + d\vec{T_3} + d\vec{T_4}$

Torque along side AB:

$\vec{T} = \vec{R} \times \vec{F}$

$d\vec{T_1} = d\vec{R_1} \times d\vec{F_1}$.

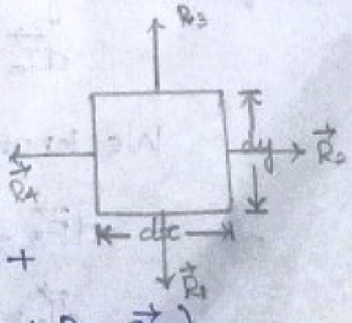
We know that:

$d\vec{F_1} = I d\vec{l_1} \times \vec{B_1}$

$d\vec{l_1} = dx\,\vec{a_x}$

$\vec{B_1} = B_{ox}\vec{a_x} + B_{oy}\vec{a_y} + B_{oz}\vec{a_z}$.

To find $\vec{R_1}$:



$\vec{R_1} = \frac{1}{2} dy(-\vec{a_y})$.

$d\vec{F_1} = I dx\,\vec{a_x} \times (B_{ox}\vec{a_x} + B_{oy}\vec{a_y} + B_{oz}\vec{a_z})$.

$d\vec{F_1} = I dx\,(B_{oy}\vec{a_z} - B_{oz}\vec{a_y})$

$d\vec{T_1} = \vec{R_1} \times d\vec{F_1}$

$$= \frac{1}{2} dy(-\vec{a_y}) \times I dx (B_{oy} \vec{a_z} - B_{oz} \vec{a_y}).$$

$$d\vec{T_1} = \frac{I dx dy}{2} (-B_{oy} \vec{a_x})$$

Torque along side BC:

$$d\vec{T_2} = \vec{R_2} \times d\vec{F_2}$$

We know that:

$$d\vec{F_2} = I d\vec{l_2} \times \vec{B_2}$$

$$d\vec{l_2} = dy \vec{a_y}.$$

$$\vec{B_2} = B_{ox} \vec{a_x} + B_{oy} \vec{a_y} + B_{oz} \vec{a_z}.$$

To find $\vec{R_2}$:

$$\vec{R_2} = \frac{1}{2} dx \vec{a_x}$$

$$d\vec{F_2} = I dy \vec{a_y} \times (B_{ox} \vec{a_x} + B_{oy} \vec{a_y} + B_{oz} \vec{a_z})$$

$$d\vec{F_2} = I dy (-B_{ox} \vec{a_z} + B_{oz} \vec{a_x}).$$

$$d\vec{T_2} = \vec{R_2} \times d\vec{F_2}.$$

$$= \frac{1}{2} dx \vec{a_x} \times (I dy (-B_{ox} \vec{a_z} + B_{oz} \vec{a_x}))$$

$$d\vec{T_2} = \frac{1}{2} I dx dy (B_{ox} \vec{a_y})$$

Torque along side CD:

$$d\vec{T_3} = \vec{R_3} \times d\vec{F_3}.$$

We know that:

$$d\vec{F_3} = I d\vec{l_3} \times \vec{B_3}.$$

$$d\vec{l_3} = dx (-\vec{a_x}).$$

$$\vec{B_3} = B_{ox} \vec{a_x} + B_{oy} \vec{a_y} + B_{oz} \vec{a_z}$$

To find $\vec{R_3}$:

$\vec{R_3} = \frac{1}{2} dy(\vec{a_y})$.

$\vec{dF_3} = I\vec{dl_3} \times \vec{B_3}$

$= Idx(-\vec{a_x}) \times (B_{0x}\vec{a_x} + B_{0y}\vec{a_y} + B_{0z}\vec{a_z})$

$\vec{dF_3} = Idx(-B_{0y}\vec{a_z} + B_{0z}\vec{a_y})$

$\vec{dT_3} = \vec{R_3} \times \vec{dF_3}$

$= \frac{1}{2} dy(\vec{a_y}) \times Idx(-B_{0y}\vec{a_z} + B_{0z}\vec{a_y})$.

$\vec{dT_3} = \frac{Idxdy}{2}(-\vec{a_x} B_{0y})$

Torque along side DA:

$\vec{dT_4} = \vec{R_4} \times \vec{dF_4}$

We know that:

$\vec{dF_4} = I\vec{dl_4} \times \vec{B_4}$.

$\vec{dl_4} = dy(-\vec{a_y})$.

$\vec{B_4} = B_{0x}\vec{a_x} + B_{0y}\vec{a_y} + B_{0z}\vec{a_z}$

To find $\vec{R_4}$:

$\vec{R_4} = \frac{1}{2} dx(-\vec{a_x})$.

$\vec{dF_4} = Idy(-\vec{a_y}) \times (B_{0x}\vec{a_x} + B_{0y}\vec{a_y} + B_{0z}\vec{a_z})$

$\vec{dF_4} = Idy(B_{0x}\vec{a_z} - B_{0z}\vec{a_x})$.

$\vec{dT_4} = \vec{R_4} \times \vec{dF_4}$

$= \frac{1}{2} dx(-\vec{a_x}) \times Idy(B_{0x}\vec{a_z} - B_{0z}\vec{a_x})$.

$\vec{dT_4} = \frac{Idxdy}{2}(B_{0x}\vec{a_y})$.

$\vec{dT} = \vec{dT_1} + \vec{dT_2} + \vec{dT_3} + \vec{dT_4}$

$$= \frac{Idxdy}{2}(-B_{oy}\vec{a_x}) + \frac{Idxdy}{2}(B_{ox}\vec{a_y})$$

$$+ \frac{Idxdy}{2}(-B_{oy}\vec{a_x}) + \frac{Idxdy}{2}(B_{ox}\vec{a_y}).$$

$$= Idxdy(-B_{oy}\vec{a_x}) + Idxdy(B_{ox}\vec{a_y}).$$

$$= Idxdy\left[B_{ox}\vec{a_y} - B_{oy}\vec{a_x}\right].$$

$$d\vec{T} = Idxdy(\vec{a_z} \times \vec{B}).$$

Surface area, $\vec{ds} = dxdy\vec{a_z}$

$$d\vec{T} = I(\vec{ds} \times \vec{B}).$$

Differential Magnetic dipole Moment,

$$d\vec{m} = I\vec{ds}$$

$$d\vec{T} = d\vec{m} \times \vec{B}.$$

$$\vec{T} = \vec{m} \times \vec{B}.$$